



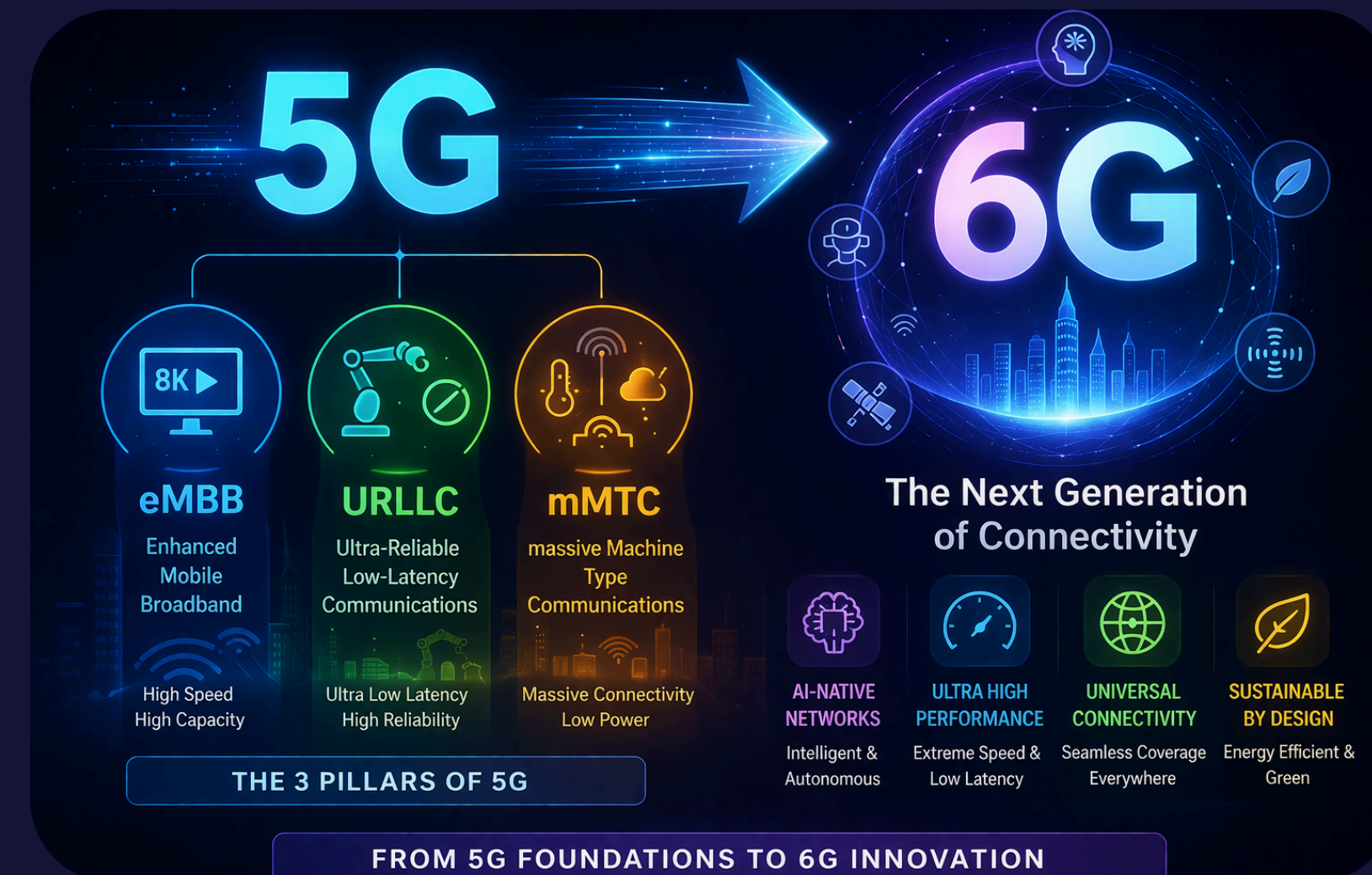
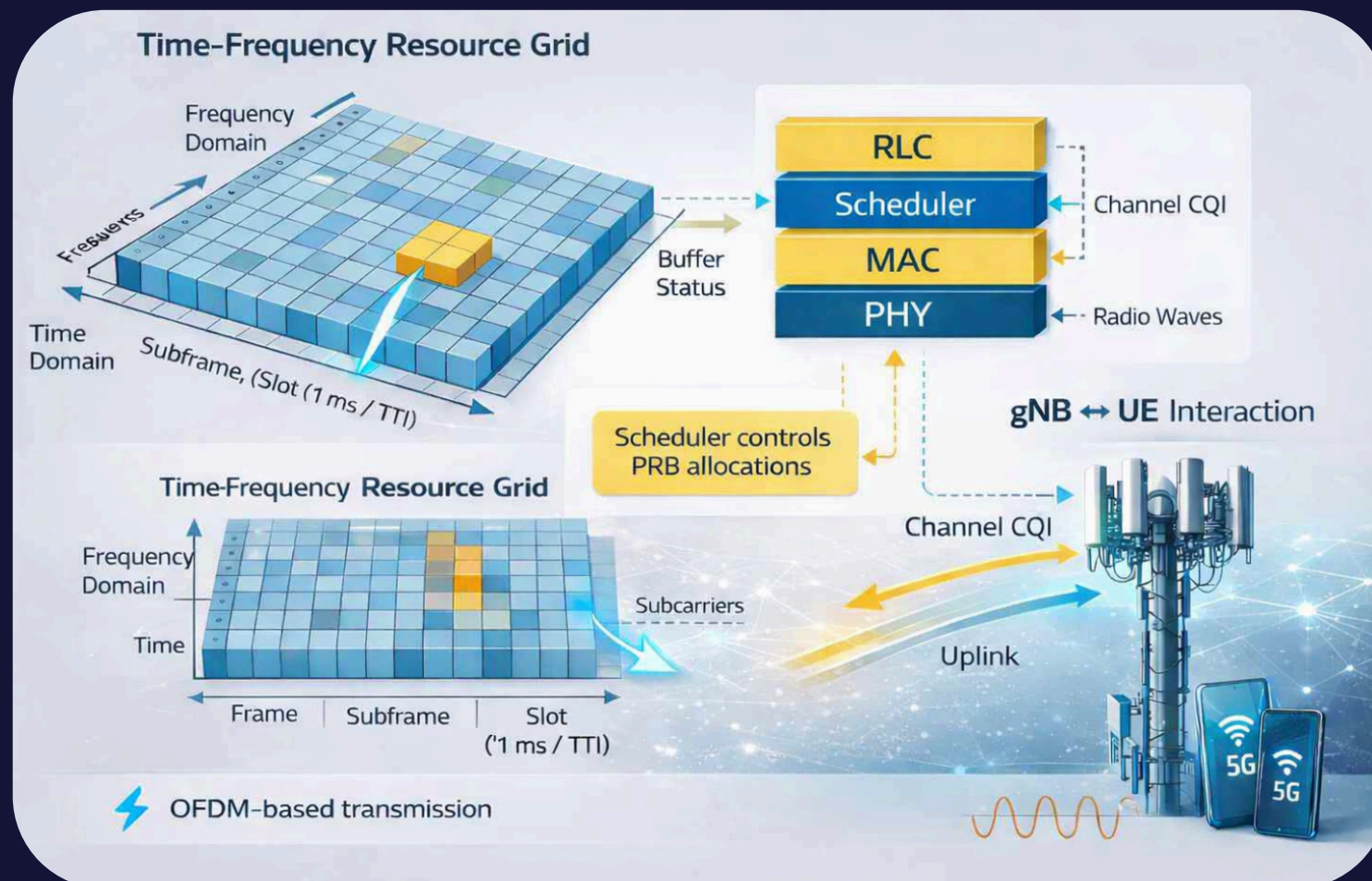
WHY SMARTER SCHEDULING IN 5G/6G

Introduction & Motivation

Evolution: LTE → 5G → 6G

- Traffic explosion:
- eMBB (high throughput)
- URLLC (low latency)
- mMTC (massive devices)

Scheduler = core decision-maker in RAN performance



Scheduling Algorithms

Max-CQI

- Schedules UE with best channel quality (highest CQI)
- High throughput
- Severe UE starvation

Round Robin (RR)

- Allocates resources equally among UEs
- Fair distribution
- Poor spectral efficiency

Proportional Fair (PF)

- Balances throughput and fairness
- Considers:
 - Instantaneous rate
 - Average throughput

Benefit:

- Favors users with good channels
- But compensates users with low past throughput

Limitations

Dynamic Network Challenges

- Rapid channel variations (CQI fluctuations)
- Sudden traffic bursts (eMBB spikes, congestion)
- Diverse QoS requirements (URLLC vs mMTC vs eMBB)

Hard-Coded Trade-offs

- Throughput vs fairness is manually designed
- No real-time optimization
- Suboptimal in complex environments

“Hence the need for intelligent schedulers that can learn and adapt.”

PF is widely adopted in LTE/5G systems as a baseline scheduler

Can AI learn better scheduling policies than handcrafted algorithms?”

AI-Based Scheduling Framework

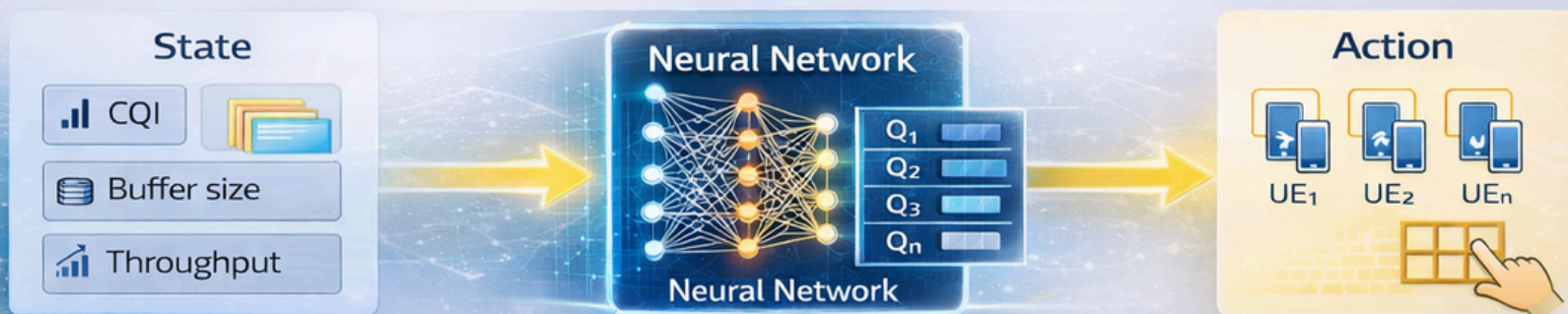
Deep Q-Network for RAN Scheduling

Model Overview

- Discrete action space
- Learns Q-values
- Neural network approximates policy

Decision Pipeline

👉 Core flow: State → Neural Network → Q-values → Action



State (Input)

- CQI (channel quality)
- Buffer size
- Average throughput
- Past allocations



Action

- Compute Priority
- Select UE
- Allocate PRBs

Reward

- Throughput ↑
- Fairness ↑

Deep RL Scheduler (DQN)

Overview

- Discrete action space
- Learns Q-values
- Neural network approximates policy

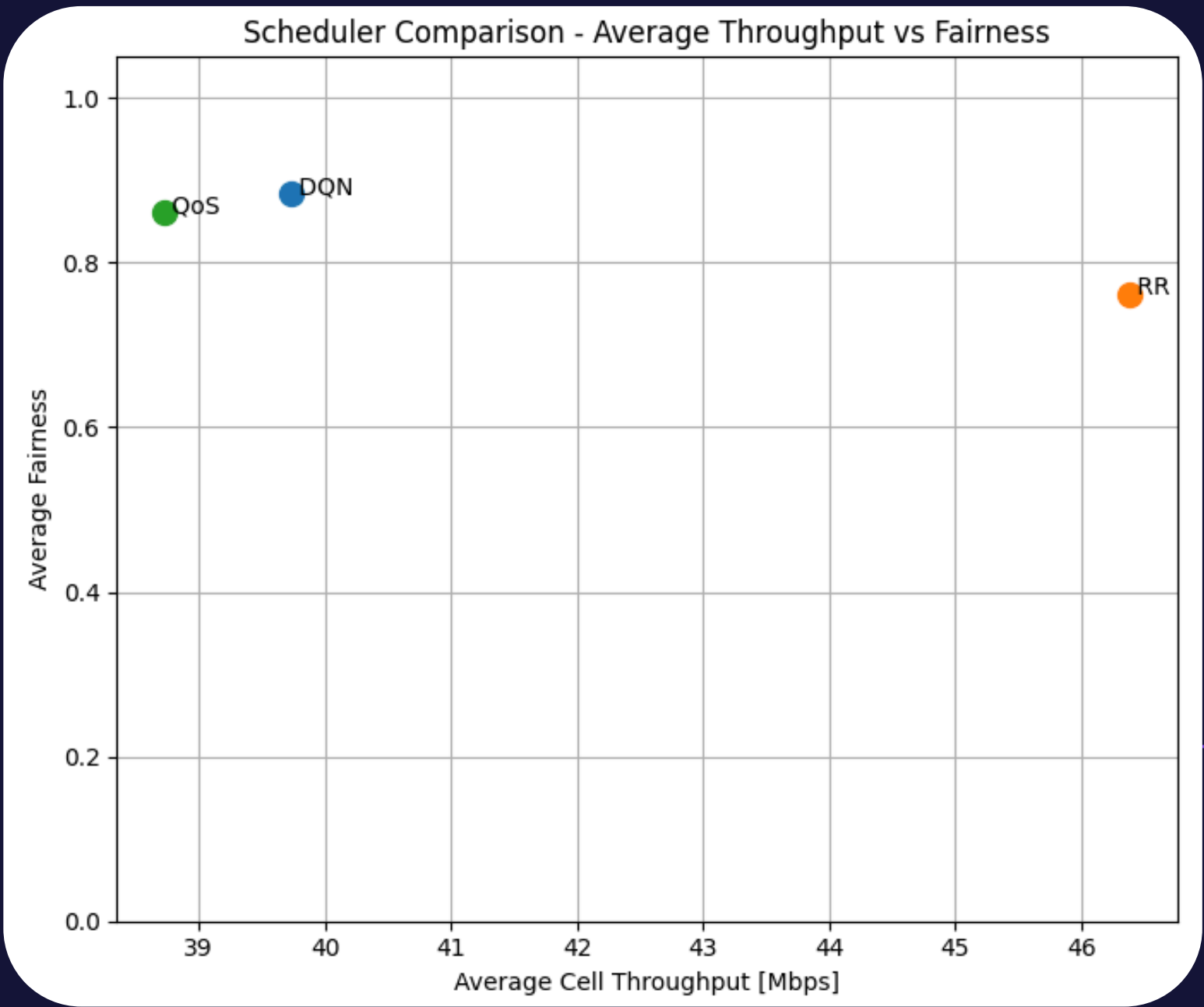
Framework

- State = network condition
- Q-value = expected reward
- Action = scheduling decision

Why DQN

- Handles complex environments
- Learns optimal policy over time
- Improves fairness & throughput

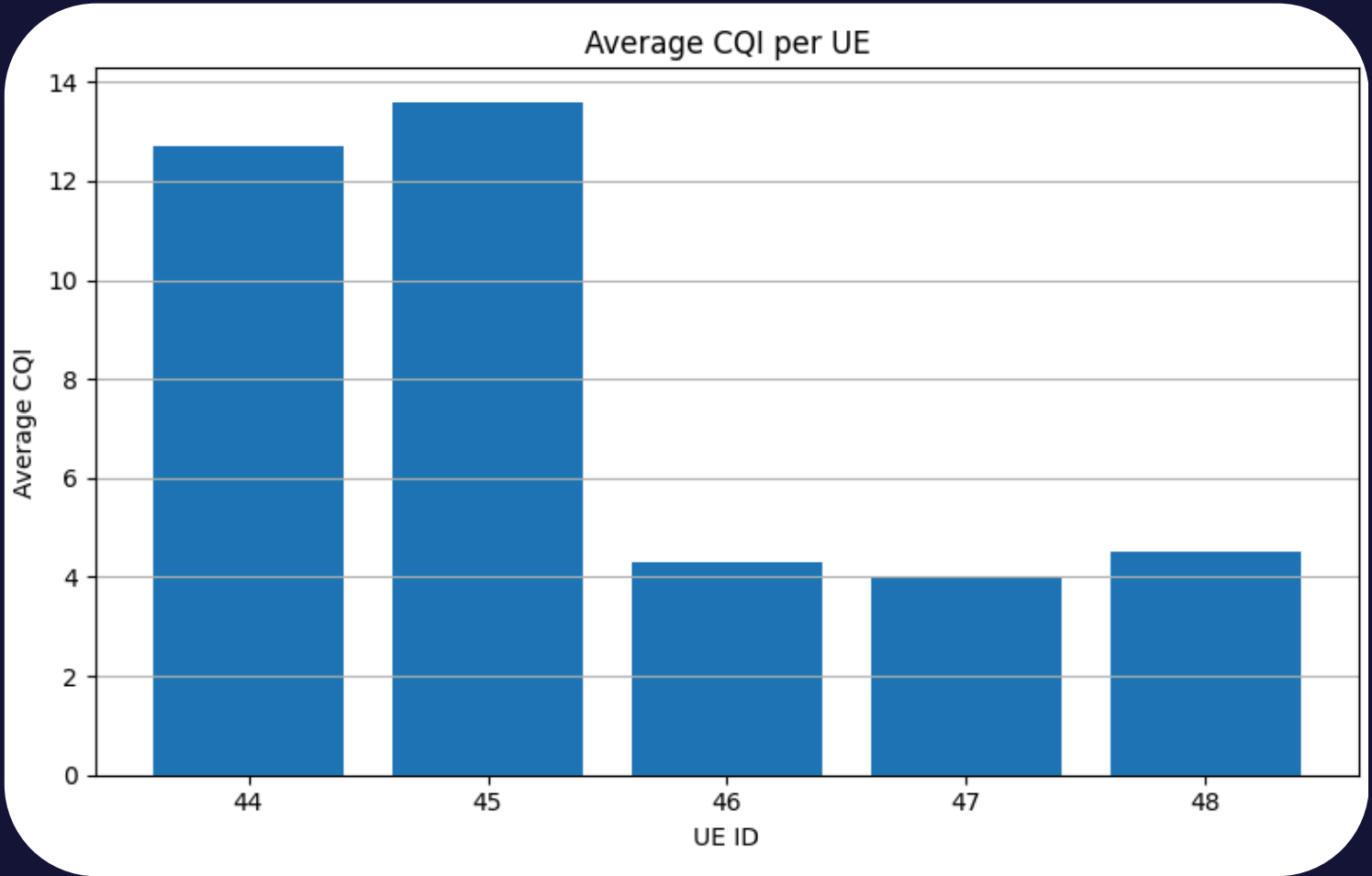
Results and Comparisons



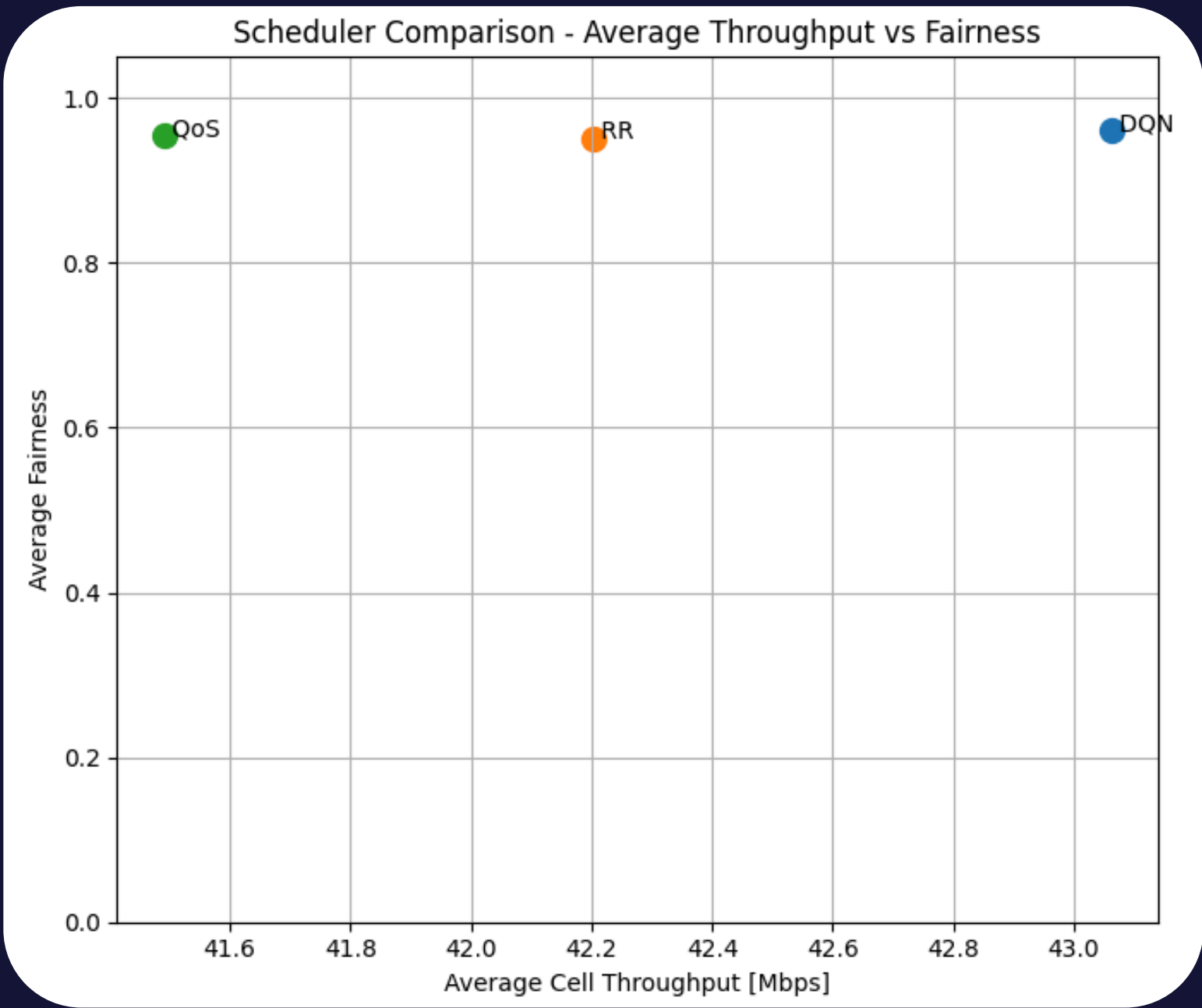
For the scenario where the average CQI per UE vastly differs, the DQN scheduler provides gain in the fairness aspect.



**MORE FAIR
SCHEDULING**

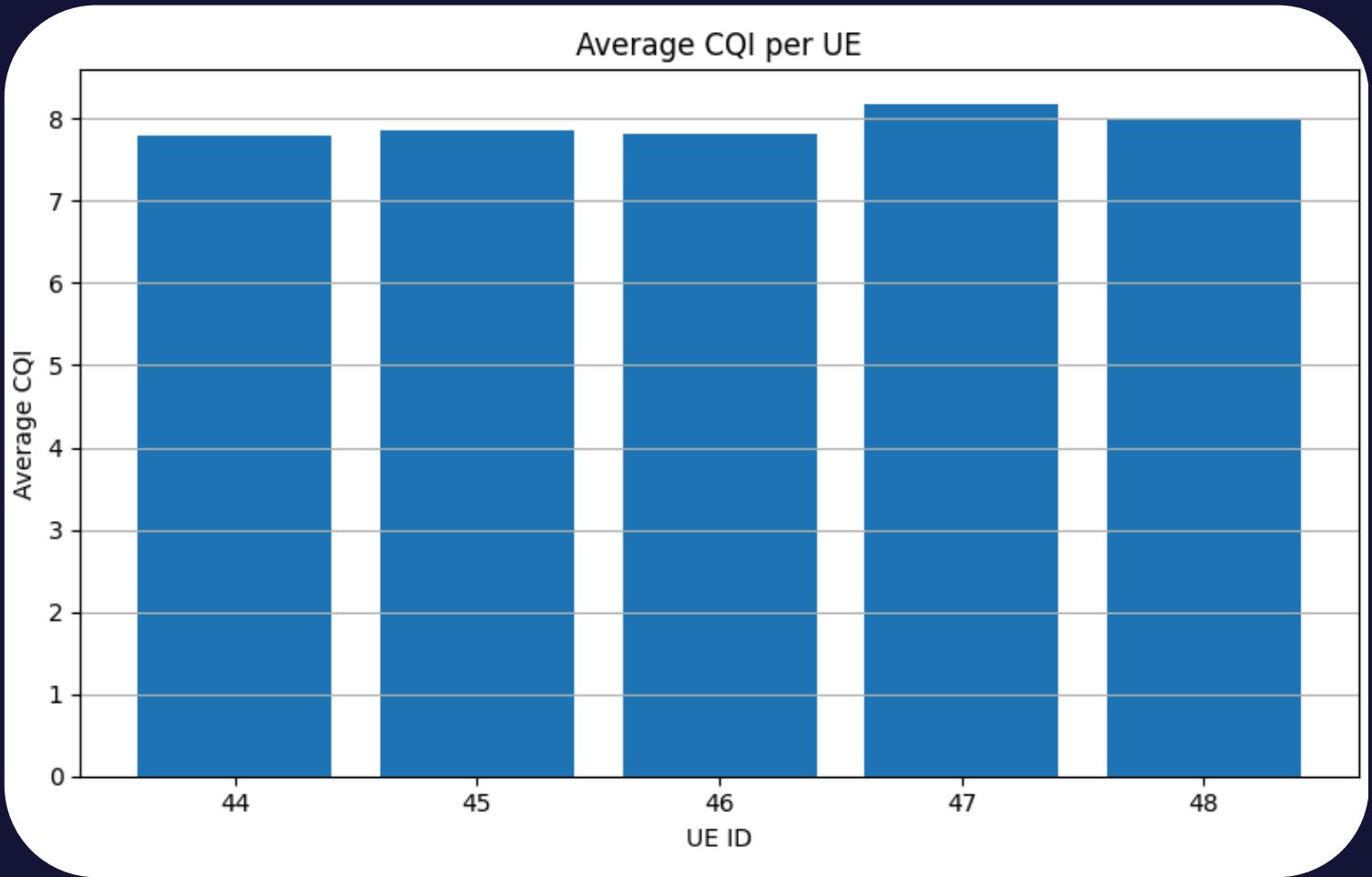


Results and Comparisons



For the scenario where the average CQI per UE difference is trivial, the DQN scheduler provides gain in the throughput aspect.


**BETTER
THROUGHPUT**



Thank you



You are welcome to interact with our dashboard and ask questions.